Software Projects Estimation (Educational Games)

Module: Early Sizing of Software Requirements

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Audience: Students

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1. Introduction/Background

This game simulates a set of measurement activities early in a software lifecycle, and game player's role is that of a manager responsible for managing this set of early sizing activities. This educational game is designed to evaluate the impact of the player's decisions on selecting the measurers, the set requirements documents and related early sizing tools selection. This simulation trains students on how to make management choices, taking into consideration several factors concurrently.

2. Learning Objectives

The three game-specific learning objectives in this game are from an early software sizing perspective:

- *The manager (e.g., the game player) needs to make decisions on a number of factors.*
    - Selection of two measurers as the measurement team.
        - ♣ All candidate measurers have different profiles, expertise, and salaries.
            - The profiles influence their accuracy in Function Points sizing.
            - The expertise influences their velocity in sizing;
            - Their salaries are related to their expertise and impact the early sizing costs.
    - Select how many software functional user requirements will be measured – leading to distinct budget according to the number of requirements.
    - Select whether a tool will be used by the measurers: the usage of a tool will influence the speed and the cost of measurement (e.g., the budget). In this SimSE game, the tool corresponds to the document: COSMIC Early Sizing Guidelines.
    - *How much budget left is not evaluated in the game.* The SimSE system game will assign to the measurers team (e.g., manager) an initial budget based on the number of requirements they selected to size. And both the team members' salaries and tool selection will cost money.
- The game player-manager must focus on completing the early sizing before spending all of the budget. As long as there is still some money left in the budget, SimSE will assign a score to the budget.

4. Prerequisites

For this game, it is not necessary to have detailed knowledge about FSM measurement rules either from COSMIC or other Function Points methods. The main objective of this game is to take management decisions on resources selection for requirements sizing activities.

5. Time Commitment

The average time to play a single game is 10-15 minutes, but, of course, it is likely to take several iterations (e.g. games) playing the game for the student to learn the concepts and be able to answer the questions. Participants should be given at least one week of out-of-class time to explore the game and answer the questions (see Section 8).

6. Instructions

In addition to the generic instructions in the first module on how to use the SimSE simulation software:

Download this game at:

> http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Games.
>
> The download consists of a "readme" text file and an executable game, which you can run by simply double-clicking on it. If you do not have the current version of Java installed on your machine, you will have the opportunity to install it when you try to run a game.

7. Questions

We recommend choosing approximately three of the following questions.

1. How can I figure out whether the two measurers selected will finish the measurement activity within the budget?
2. Why is the average budget per requirement for the set of 50 requirements lower than the average per requirement for the set of 10 requirements?
3. What is a measurer's salary based on?
4. What will be the measurement impact in term of measurement speed and measurement error when:
    a. one of the measurers selected has a perfectionist profile and a minor level expertise in measurement, and
    b. the other measurer has an expeditious profile but a major level of expertise in measurement?

8. Feedback?

If you have any comments, suggestions, feedback, or experience regarding this course module or SimSE in general, please send an email to Alain Abran<Alain.Abran@etsmtl.ca>

References

1. Navarro, E.O. *"The Fundamental Rules" of Software Engineering.* 2008 [Available from: http://www.ics.uci.edu/~emilyo/SimSE/se_rules.html].

2. *COSMIC. (2020). Early Software Sizing with COSMIC: Practitioners Guide.*

3. *COSMIC. (2020). Early-Sizing-Experts-Guide-May-2020-1. https://doi.org/10.13140/RG.2.1.4195.0567*

4. *Abran, A. (2015).* Software project estimation: The fundamentals for providing high quality information to decision makers. *John Wiley & Sons.*